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Nishant Joshi

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PERSONAL SUMMARY

Computational neuroscientist with expertise in large-scale reproducible Python pipelines, signal processing, and machine learning for high-dimensional electrophysiological data. Focused on building interpretable, biologically grounded models of neural computation. Interested in translational electrophysiology, sleep neurophysiology, and closed-loop neuromodulation.

RESEARCH EXPERIENCE

PhD Candidate (Thesis approved)

05/2021 — 12/2025

Donders Institute for Brain, Cognition and Behavior | Radboud University, NL

Prof. Dr. Tansu Celikel (Georgia Tech, USA) and Dr. Fleur Zeldenrust (Radboud, NL)

Nijmegen, Netherlands

- Designed and deployed a scalable end-to-end analysis pipeline for large-scale, high-dimensional electrophysiological data (3.6+ billion data points from 1200+ recordings), enabling multimodal feature extraction, integration, and modeling.
- Designed heterogeneous-timescale reservoir and recurrent network architectures, in collaboration with Ghent University, translating single-neuron timescale diversity into a network-level design principle; improved memory capacity and performance on NARMA and Mackey-Glass benchmarks over standard reservoirs.
- Spearheaded collaborations with the Netherlands eScience Center, applying FAIR-compliant data management methods and advanced machine-learning techniques to biological data.
- Actively participated in scientific dissemination through conference posters and abstracts. Also, designed course material for the Dutch Brain Bee and carried out outreach activities.

MSCA Visiting Researcher

02/2024 — 04/2024

Prof. Dr. Panayiota Poirazi (IMBB Forth, GR)

Iraklion, Greece

- As part of the MSCA ITN, I developed a pipeline for fitting generalized leaky integrate-and-fire models to single-unit electrophysiological data.
- Developed analysis workflows to incorporate the passive features extracted from the model into unsupervised clustering results.

Research Software Engineer (MSCA Secondment)

02/2023 — 05/2023

Netherlands eScience Center

Amsterdam, Netherlands

- Designed reproducibility-focused machine learning workflows for unsupervised clustering and integrative analysis of high-dimensional biological data, using Git, CI/CD, Docker, and FAIR data principles.
- Contributed to a long-term collaboration between the Netherlands eScience Center and the Donders Center for Neuroscience, resulting in joint publications and conference abstracts.

Research Assistant

02/2020 — 04/2021

Prof. Dr. Abigail Morrison (INM6/IAS6 – Forschungszentrum Jülich, DE)

Jülich, Germany

- Developed scalable pipelines for designing, training, and analyzing more than 100 recurrent and spiking neural networks on GPU-enabled JURECA Supercomputer HPC infrastructure.
- Applied fixed-point analysis, dimensionality reduction, and representational similarity analysis to study model behavior across architectures and hyperparameter settings.
- Managed publication library for INM-6 and designed study material on GANs for a seminar course at RWTH Aachen.

EDUCATION

PhD – Computational Neuroscience (thesis approved) <i>Radboud University, Nijmegen, NL</i>	05/2021 – 12/2025
Master of Science – Computer Science and Engineering (Dual Degree) <i>KTH Royal Institute of Technology, Stockholm, SE & TU Berlin, Berlin, DE (Grade 9.1/10, Thesis: 10/10)</i>	10/2018 – 11/2020
Bachelor of Engineering – Industrial Engineering <i>R.V. College of Engineering, Bangalore, IN (Grade: First Class with Distinction)</i>	08/2013 – 05/2017

PUBLICATIONS

- **Joshi, N.**, van Der Burg, S., Celikel, T., & Zeldenrust, F. (2025). Neuronal identity is not static: An input-driven perspective. PLOS Computational Biology, 21(12), e1013821.
 - **Joshi, N.**, Yan, X., ..., Celikel, T., & Zeldenrust, F. (2026). Neuromodulatory Control of Cortical Function: Cell-Type Specific Regulation of Neuronal Information Transfer. bioRxiv, 2026-03. (Under revision at PNAS)
- Manuscripts in preparation**
- Iacob[†] S, **Joshi[†] N**, Dambre J, Zeldenrust F. Timescale Heterogeneity in Reservoir Computing
- Dataset**
- Joshi, N., Celikel, T., Zeldenrust,..., Calcini, N., Neuromodulatory influence on information transfer of single cortical neurons, <https://doi.org/10.34973/nhes-qe97> (to be released upon publication)
- Posters**
- Iacob, S.[†], **Joshi, N.[†]**, Dambre, J., Zeldenrust, F. (2024). Co-evolved structural and temporal network heterogeneity. Bernstein Conference 2024. <https://doi.org/10.12751/nncn.bc2024.309>
 - **Joshi, N.**, Celikel, T., Zeldenrust, F. (2023). High dimensional clustering reveals heterogeneity in excitatory and inhibitory cells in the neocortex. Bernstein Conference 2023. <https://doi.org/10.12751/nncn.bc2023.095>

SKILLS

Modeling & Analysis	Python (advanced), MATLAB (signal processing), statistical modeling, dimensionality reduction, clustering, probabilistic modeling, multimodal data integration, high-dimensional biological data analysis, time-series analysis
Machine Learning Framework	TensorFlow (SNN, RNN, VAE, Siamese Networks, GANs), PyTorch and Keras
Infrastructure & Tools	Git, Docker, Linux, HPC (SLURM, Horovod, CUDA), CI/CD, FAIR data principles, LaTeX, Kubernetes

OUTREACH AND TEACHING ACTIVITIES

- Supervised 4 students (2 bachelor and 2 master)
- Presented my research in the form of posters at 4 international conferences
- Designed Course Material for Dutch Brain Bee and TA for Mathematics for Biology
- Presented my methods at AI for Science symposium (Radboud University)
- Attended workshops organized by the MSCA ITN network at TU Munich, Uni-Konstanz, Uni-Ghent, Radboud and Uni-Crete

ACHIEVEMENTS

- Marie Skłodowska Curie ITN fellowship, 2021 – 2024
- EIT Digital Excellence Scholarship, 2018 – 2020
- Best Thesis Award, Department of Industrial Engineering, 2017